WORKED SOLUTION

STEP II 2010, Question 2

Pure • Trigonometric Integration

STEP II 2010, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q2

The key idea

The cubic $7\sin x - 8\sin^3 x$ looks awkward to integrate until we notice that it is built from the triple-angle formula: twice $\sin 3x$ plus a bit left over. The triple-angle formula for cosine then turns the answer back into a polynomial in $c = \cos\alpha$. For Eustace's integral, $\sin^2\alpha = 1 - c^2$ puts his (wrong) answer into the same variable, and comparing the two gives a quartic in c whose roots we have been handed a hint about.

Solution

The triple-angle formulae

Using the addition formulae and $\sin^2 x = 1 - \cos^2 x$,

$$\begin{aligned}\cos 3x = \cos(2x + x) &= \cos 2x \cos x - \sin 2x \sin x \\ &= (2\cos^2 x - 1)\cos x - 2\sin^2 x \cos x \\ &= 2\cos^3 x - \cos x - 2(1 - \cos^2 x)\cos x \\ &= 4\cos^3 x - 3\cos x,\end{aligned}$$

as required. In the same way, using $\cos^2 x = 1 - \sin^2 x$,

$$\begin{aligned}\sin 3x = \sin(2x + x) &= \sin 2x \cos x + \cos 2x \sin x \\ &= 2\sin x \cos^2 x + (1 - 2\sin^2 x)\sin x \\ &= 2\sin x(1 - \sin^2 x) + \sin x - 2\sin^3 x \\ &= 3\sin x - 4\sin^3 x.\end{aligned}$$

Part (i)

From the sine formula, $8\sin^3 x = 6\sin x - 2\sin 3x$, so

$$7\sin x - 8\sin^3 x = \sin x + 2\sin 3x.$$

Now the integral is easy:

$$\mathrm{I}(\alpha) = \Big[-\cos x - \tfrac{2}{3}\cos 3x \Big]_0^{\alpha} = -\cos\alpha - \tfrac{2}{3}\cos 3\alpha + 1 + \tfrac{2}{3}.$$

By the cosine formula, $\cos 3\alpha = 4c^3 - 3c$, so

$$\mathrm{I}(\alpha) = -c - \tfrac{2}{3}(4c^3 - 3c) + \tfrac{5}{3} = -\tfrac{8}{3}c^3 + c + \tfrac{5}{3},$$

which is the result we wanted.

For a value of c giving $\mathrm{I}(\alpha) = 0$: taking $\alpha = 0$ makes the integral zero trivially, so $c = 1$ works (check: $-\frac{8}{3} + 1 + \frac{5}{3} = 0$). In fact it is the only one, because $8c^3 - 3c - 5 = (c-1)(8c^2 + 8c + 5)$ and the quadratic has discriminant $64 - 160 < 0$.

Part (ii)

Eustace's rule gives $\int \sin x \,\mathrm{d}x = \frac{1}{2}\sin^2 x$ and $\int \sin^3 x \,\mathrm{d}x = \frac{1}{4}\sin^4 x$, so his value of the definite integral is

$$\mathrm{E}(\alpha) = \Big[\tfrac{7}{2}\sin^2 x - 2\sin^4 x \Big]_0^{\alpha} = \tfrac{7}{2}s^2 - 2s^4, \qquad s = \sin\alpha.$$

To compare with part (i), write $s^2 = 1 - c^2$:

$$\mathrm{E}(\alpha) = \tfrac{7}{2}(1 - c^2) - 2(1 - c^2)^2 = \tfrac{3}{2} + \tfrac{1}{2}c^2 - 2c^4.$$

The value $\cos\beta = -\frac{1}{6}$. With $c = -\frac{1}{6}$,

$$\mathrm{I}(\beta) = -\tfrac{8}{3}\cdot(-\tfrac{1}{216}) - \tfrac{1}{6} + \tfrac{5}{3} = \tfrac{1}{81} + \tfrac{3}{2} = \tfrac{245}{162},$$
$$\mathrm{E}(\beta) = \tfrac{3}{2} + \tfrac{1}{2}\cdot\tfrac{1}{36} - 2\cdot\tfrac{1}{1296} = \tfrac{3}{2} + \tfrac{1}{72} - \tfrac{1}{648} = \tfrac{3}{2} + \tfrac{8}{648} = \tfrac{245}{162}.$$

The two agree, so Eustace gets the right value of $\mathrm{I}(\beta)$ by luck.

All values of α. Eustace is right exactly when $\mathrm{E}(\alpha) = \mathrm{I}(\alpha)$, that is (multiplying by 6 and collecting terms)

$$12c^4 - 16c^3 - 3c^2 + 6c + 1 = 0.$$

? **Ask yourself** How do we factorise a quartic like this without a formula?

We already know two roots. The value $c = -\frac{1}{6}$ works, by the calculation just done, so $6c + 1$ is a factor. And $c = 1$ (that is, $\alpha = 0$) must work, because both I and E are 0 there. Dividing out $6c + 1$ (compare coefficients of c^3 and c),

$$12c^4 - 16c^3 - 3c^2 + 6c + 1 = (6c + 1)(2c^3 - 3c^2 + 1),$$

and then, since $c = 1$ is a root of the cubic,

$$2c^3 - 3c^2 + 1 = (c - 1)(2c^2 - c - 1) = (c - 1)^2(2c + 1).$$

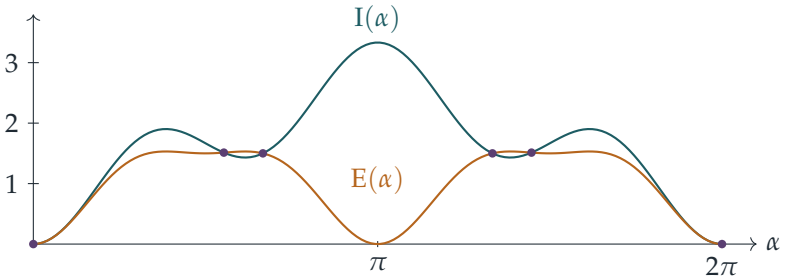
So the condition is

$$(6c+1)(2c+1)(c-1)^2 = 0, \qquad \text{that is} \qquad c = 1,\ -\tfrac{1}{2} \text{ or } -\tfrac{1}{6}.$$

Since $c = \cos\alpha$ and every value of c in $[-1,1]$ comes from some α, Eustace gets the correct value of $\mathrm{I}(\alpha)$ exactly when, for some integer n,

$$\alpha = 2n\pi, \qquad \alpha = 2n\pi \pm \tfrac{2\pi}{3}, \qquad \text{or} \qquad \alpha = 2n\pi \pm \arccos\left(-\tfrac{1}{6}\right).$$

(Negative α is included: both I and E depend on α only through $\cos\alpha$.) The graph shows the true integral (teal) and Eustace's answer (amber) for $0 \le \alpha \le 2\pi$; they meet at $\alpha = 0$, $\frac{2\pi}{3}$, $\arccos(-\frac{1}{6}) \approx 1.738$, and the mirror images of these in $\alpha = \pi$.



Checks

- At $\alpha = \pi$ ($c = -1$) the formula gives $\mathrm{I}(\pi) = \frac{8}{3} - 1 + \frac{5}{3} = \frac{10}{3}$, which matches $\int_0^\pi (\sin x + 2\sin 3x)\,\mathrm{d}x = 2 + \frac{4}{3}$.
- Simpson's rule on the original integrand gives $\mathrm{I}(\beta) \approx 1.5123457$, matching $\frac{245}{162} = 1.5123456\ldots$; at $\alpha = \frac{2\pi}{3}$ both the integral and Eustace's answer equal 1.5.

Common slips

- Forgetting the lower limit: $-\cos 0 - \frac{2}{3}\cos 0 = -\frac{5}{3}$, which supplies the $+\frac{5}{3}$.
- Missing that $c = 1$ is a *double* root of the quartic, or forgetting it altogether and reporting only $c = -\frac{1}{2}$ and $c = -\frac{1}{6}$.
- Giving only α in $[0, \pi]$; the question asks for all values.

Beyond STEP

Writing powers like $\sin^3 x$ as combinations of $\sin x$ and $\sin 3x$ is the start of Fourier analysis: every power $\sin^n x$ is a finite sum of $\sin kx$ or $\cos kx$, which you can find quickly using $\sin x = (\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x})/2\mathrm{i}$. The polynomial expressing $\cos nx$ in terms of $\cos x$ is the Chebyshev polynomial T_n, with $T_3(c) = 4c^3 - 3c$. Needham's *Visual Complex Analysis* shows why de Moivre's theorem makes all these identities natural.

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